QUANTUM MECHANICS

20/6/23

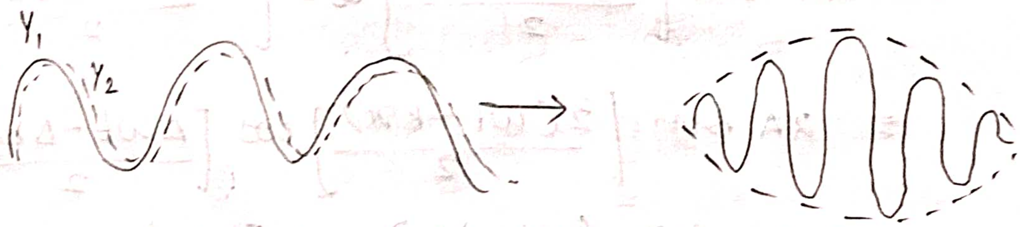
(V_P) **Phase Velocity** :- It is defined as the rate of advancement of a point at constant phase along the propagation of a wave.

$$V_P = \frac{\omega}{K}$$

$\omega \rightarrow 2\pi f$ = angular frequency

$k = \frac{2\pi}{\lambda}$ = wave vector

(V_g) **Group Velocity** :- It is the velocity with which the envelope enclosing a wave group formed due to super position of 2 or more waves propagated



consider two waves of same amplitude but slightly different wavelengths propagating along x-direction. Let y_1 and y_2 be the displacements of two waves such that

$$Y_1 = A \sin(\omega t - kx) \longrightarrow ①$$

$$Y_2 = A \sin((\omega + \Delta\omega)t - (k + \Delta k)x]$$

Where $\Delta\omega$ & Δk are small increments to omega and k respectively.

Let Y be the displacement of the resultant wave.

from the principal of super position

$$Y = Y_1 + Y_2$$

$$Y = A \sin(\omega t - kx) + A \sin[(\omega + \Delta\omega)t - (k + \Delta k)x]$$

multiply & Divide by 2.

$$Y = 2A \sin \frac{[(\omega t - kx) + (\omega + \Delta\omega)t - (k + \Delta k)]x]}{2} \times$$

$$\cos\left[\frac{(\omega t - kx) - [(\omega + \Delta\omega)t - (k + \Delta k)x]}{2}\right]$$

$$= 2A \sin\left[\frac{\omega t - kx + \omega t + \Delta t\omega - kx + \Delta kx}{2}\right] \cdot \cos\left[\frac{\omega t - kx - [\omega t + \Delta\omega t - kx - \Delta kx]}{2}\right]$$

$$= 2A \sin\left[\frac{(2\omega t - \Delta\omega t) + (2kx - \Delta kx)}{2}\right]\left[\cos\left[\frac{\omega t - kx - \omega t - \Delta\omega t + kx + \Delta kx}{2}\right]\right.$$

$$= 2A \sin\left[\frac{(2\omega + \Delta\omega)t - (2k + \Delta k)x}{2}\right] \cos\left[\frac{-\Delta\omega t + \Delta kx}{2}\right]$$

Since $\Delta\omega$ & Δk are small.

$2\omega + \Delta\omega \approx 2\omega \qquad 2k + \Delta k \approx 2k.$

$$y = 2A \sin\left[\frac{2\omega t - 2kx}{2}\right] \cos\left[\frac{\Delta\omega t - \Delta kx}{2}\right]$$

$$= 2A \sin\left[\frac{2(\omega t - kx)}{2}\right] \cos\left[\frac{\Delta\omega t - \Delta kx}{2}\right]$$

$$= 2A \sin\, \cancel{\sin} (\omega t - kx) \cos\left[\frac{\Delta\omega t - \Delta kx}{2}\right]$$

$$\Rightarrow y = 2A \cos\left(\frac{\Delta\omega t - \Delta kx}{2}\right) \sin(\omega t - kx) \longrightarrow (2)$$

comparing ① & ②.

Resultant amplitude $= 2A \cos\left(\frac{\Delta\omega t - \Delta kx}{2}\right)$

$2A \neq 0 \qquad \cos\left(\frac{\Delta\omega t - \Delta kx}{2}\right) = 0$

$$\frac{\Delta\omega t - \Delta kx}{2} = 0.$$

$$\frac{\Delta\omega t}{2} = \frac{\Delta kx}{2}$$

$$\Delta\omega t = \Delta kx.$$

$$V_g = \frac{x}{t} = \frac{\Delta\omega}{\Delta t} \qquad \boxed{V_g = \frac{d\omega}{dk}}$$

* RELATION BTN PHASE VELOCITY & GROUP VELOCITY

w.k.t $V_g = \dfrac{d\omega}{dk} \rightarrow ①$.

Also $V_p = \dfrac{\omega}{k}$.

$\therefore \omega = V_p \cdot k \rightarrow ②$

Subs ② in ① we get;

$$V_g = \frac{d}{dk}(V_p k).$$

$$V_g = V_p \cdot \frac{dk}{dk} + k \cdot \frac{dV_p}{dk}.$$

but $k = 2\pi/\lambda$

$$V_g = V_p + \frac{2\pi}{\lambda}\left(\frac{dV_p}{d(2\pi/\lambda)}\right).$$

$$= V_p + \frac{2\pi}{\lambda}\left(\frac{dV_p}{2\pi d(1/\lambda)}\right).$$

$$\frac{1}{\lambda} = \frac{\lambda}{\lambda^2}$$

$$V_g = V_p + \frac{2\pi}{\lambda} \cdot \frac{dV_p}{2\pi \frac{d\lambda}{\lambda^2}}$$

$$= V_p - \frac{\lambda^2}{\lambda}\left(\frac{dV_p}{d\lambda}\right)$$

$$\boxed{V_g = V_p - \lambda \frac{dV_p}{d\lambda}}$$ For dispersive medium

For Non-dispersive medium

$$\frac{dV_p}{d\lambda} = 0. \qquad \boxed{\therefore V_g = V_p}$$

$V_p = \dfrac{\omega}{k}$. $\quad \omega = 2\pi\nu$

$\omega = \dfrac{2\pi E}{h}$

$\lambda = \dfrac{h}{p}$ $\quad k = \dfrac{2\pi}{\lambda}$

$\dfrac{1}{\lambda} = p/h$ $\quad k = \dfrac{2\pi p}{h}$.

$$V_p = \frac{E}{p} = \frac{mc^2}{mv}$$

$$V_p = \frac{c^2}{V_g}.$$

Applications ... NON EXISTANCE OF ELECTRON MADE THE NUCLEUS.

consider the nucleus of an atom with diameter 10^{-14} m. The uncertainity on locating the position of an electron inside the nucleus = the diameter of the nucleus itself

$$\therefore d = \Delta x = 10^{-14} m$$

From uncertainity principle,

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta p \simeq \Delta \geq \frac{h}{4\pi \Delta x}$$

$\boxed{h = 6.63 \times 10^{-34}}$

$$p = \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 10^{-14}}$$

$$p = 0.5 \times 10^{-20} \text{ Ns}.$$

The energy of electron,

$$\boxed{E = mc^2}$$

$$\boxed{\text{But } m = \frac{m_0}{\sqrt{1 - V^2/c^2}}}$$

① Energy $= mc^2$.
② Momentum(p) $= mv$

$$E = \frac{m_0 c^2}{\sqrt{1 - V^2/c^2}}$$

$$E^2 = \frac{m_0^2 c^4}{\frac{c^2 - V^2}{c^2}} = \frac{m_0^2 c^6}{c^2 - V^2} \rightarrow ②.$$

The momentum $p = mv$

$$p = \frac{m_0 V}{\sqrt{1 - V^2/c^2}}$$

$$p^2 = \frac{m_0^2 V^2}{1 - V^2/c^2} = \frac{m_0^2 V^2}{\frac{c^2 - V^2}{c^2}}$$

$$p^2 = \frac{m_0^2 V^2 c^2}{c^2 - V^2}$$

$$\times c^2 \quad p^2 c^2 = \frac{m_0^2 c^4 V^2}{c^2 - V^2} \rightarrow ③$$

Eqn (2) - (3), we get

$$E^2 - p^2c^2 = \frac{m_0^2c^6}{c^2-v^2} - \frac{m_0^2c^4v^2}{c^2-v^2}$$

$$E^2 - p^2c^2 = \frac{m_0^2c^4}{\cancel{c^2-v^2}}\cancel{[c^2-v^2]}$$

$$E^2 - p^2c^2 = m_0^2c^4$$

$$E^2 = p^2c^2 + m_0^2c^4$$

$$E = \sqrt{p^2c^2 + m_0^2c^4}$$

$$E = \sqrt{(0.5\times10^{-20})^2\times(3\times10^8)^2 + (9.1\times10^{-31})^2(3\times10^8)^4}$$

$$E = 1.5\times10^{-12}\ J$$

$$\text{or } \frac{1.5\times10^{-12}}{1.6\times10^{-19}}\ eV$$

$$E = 9.4\times10^6\ eV$$

$$\boxed{E = 9.4\ MeV}$$

Conclusion :- If electron exist within the nucleus then its energy will be 9.4 MeV. But β decay experiments shows that the kinetic energy of the electrons ejected during β decay or above 3-4 MeV. Hence we can conclude that electrons cannot exist within the nucleus.

* HEISENBERG'S UNCERTAINITY PRINCIPLE :-

↳ [State that it is impossible to measure both position and momentum of a particle simultaneously and accurately] If (Δx) represents uncertainity in the measurment of a particle and (Δp) represents uncertainity in the measurment of its momentum then according to uncertainity principle

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

The uncertainity principle can also be extended to connonical conjugates such as energy and time, angular displacement and angular momentum

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

$$\Delta\theta \cdot \Delta L \geq \frac{h}{4\pi}$$

According to uncertainity principle, if the position of a particle is known accurately then its momentum will be unknown or vice versa.

$\Delta x \to 0$. $\quad \Delta x \geq \frac{h}{4\pi \Delta p}$

$\Delta p \to \infty$

or $\Delta p \to 0 \quad \Delta p \geq \frac{h}{4\pi \Delta x}$.

$\Delta x \to \infty$

* Schrodinger's Time Independent Wave Equation:

Consider a particle moving along x direction. The wave function for the particle is assumed to be the soln of an undamped oscillator.

$$\boxed{\psi(x,t) = \psi_0 e^{-i\omega t}} \to (1)$$

. where $\psi_0 \to$ amplitude

The differential wave eqn for this particle is

$$\boxed{\frac{d^2\psi}{dx^2} = \frac{1}{u^2}\frac{d^2\psi}{dt^2}} \to (2)$$

. $u \to$ wave velocity.

diff (1) wrt 't'

$$\frac{d\psi}{dt} = \psi_0(-i\omega)e^{-i\omega t}$$

$$\frac{d\psi}{dt} = -i\omega\psi_0 e^{-\omega t}$$

diff again wrt t.

$$\frac{d^2\omega}{dt^2} = i\omega\psi_0(-i\omega)e^{-i\omega t}$$

$$\frac{d^2\psi}{dt^2} = i^2\omega^2\psi_0 e^{-i\omega t}$$

$$\frac{d^2\psi}{dt^2} = -\omega^2\psi_0 e^{-i\omega t}$$

$$\frac{d^2\psi}{dt^2} = -\omega^2\psi \to (3).$$

$[\therefore \psi = \psi_0 e^{-i\omega t}]$.

Sub (3) in (2) we get.

$$\boxed{\frac{d^2\psi}{dx^2} = \frac{-\omega^2}{u^2}\psi}$$

but $\omega = 2\pi f \quad u = f\lambda$

$$\frac{d^2\psi}{dx^2} = \frac{(2\pi f)^2}{(f\lambda)^2}\psi$$

$$= \frac{4\pi^2 f^2}{f^2\lambda^2}\psi$$

$$\frac{d^2\psi}{dx^2} = -\frac{4\pi^2}{\lambda^2}\psi$$

$$\frac{d^2\psi}{dx^2} = \frac{-4\pi^2}{\frac{h^2}{m^2v^2}}\psi \qquad \boxed{\lambda = \frac{h}{mv}}$$

$$\frac{d^2\psi}{dx^2} = \frac{-4\pi^2 m^2 v^2}{h^2}\psi \to (4).$$

let 'E' be the total energy and V be the P.E of particle.

$KE = E - V$.

$\frac{1}{2}mv^2 = [E-V]$.

$\times$ ly $\frac{1}{2}m^2v^2 = m[E-V]$

$m^2v^2 = 2m[E-V]$

Sub m^2v^2 in (4) we get

$$\frac{d^2\psi}{dx^2} = \frac{-4\pi^2 \, 2m(E-V)}{h^2}\psi$$

$$\frac{d^2\psi}{dx^2} = \frac{-8\pi^2 m}{h^2}[E-V]\psi$$

$$\boxed{\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}[E-V]\psi = 0}$$

5m

* WAVE FUNCTION :-

According to Man Born, all the quantum particles are associated with a wave character. Hence a wave function provides all possible information about the wave character associated with the particle.

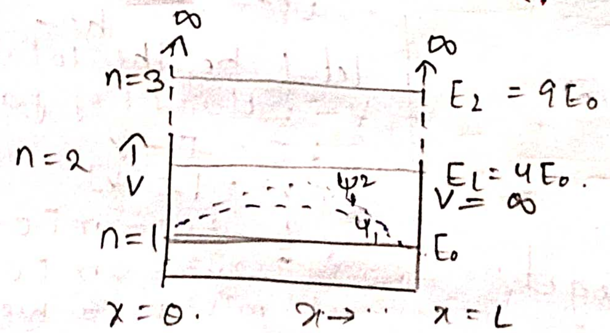
The wave function (ψ) do not have any physical significance, wheras ψ^2 has a significance. It represents the probility of finding the particle in a given elemental volume. Therefore the probility of finding the particle in a given volume at a given instant of time is given by $P = \int_{-\infty}^{\infty} |\psi|^2 dv = 1$

The value of probability lies btn 0 and 1.
1 corresponds to certainity of its presence and
0 corresponds to certainty of its absence.

The wave function has to satisfy the following properties or characteristics to find the probability. They are :-

(i) ψ must be single value.
(ii) ψ must be finite everywhere.
(iii) ψ and its first derivative are continous.

* PARTICLE IN AN INFINITE POTENTIAL WELL OR PARTICLE IN A BOX.



$\psi = 0 \quad 0 \leq x$

$\psi = 0 \quad x \geq L$

consider a particle confined to potential wealth of infinite height. Let the particle be moving between the two perfectly rigid walls hence particle confined within the potential wealth. The potential energy of the particle inside the well is assumed to be zero.

The schrodinger's wave eqn for particle inside the well :-

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}(E-V)\psi = 0.$$

$$\frac{d^2\psi}{dx^2} + \frac{8\pi^2 m}{h^2}E\psi = 0. \quad \Big| \therefore V=0.$$

$$\frac{d^2\psi}{dx^2} + k^2\psi = 0 \rightarrow ①.$$

where $k^2 = \frac{8\pi^2 m}{h^2}E$

The soln of eqn ① SDE is

$$\psi = A\sin kx + B\cos kx \rightarrow ②$$

where, A and B are constants

consider boundary condition ①

$\psi = 0$ at $x = 0$.

$$② \Rightarrow 0 = A \sin^{\nearrow 0} k(0) + B\cos k(0)$$

$$\therefore \underline{B = 0}.$$

consider Boundary condition ②.

$\psi = 0$ at $x = L$

$$② \Rightarrow 0 = A\sin KL + B\cos KL$$

$$0 = A\sin KL + 0 \cos^{\nearrow 0} KL \qquad B=0.$$

$$A\sin KL = 0.$$

$A \neq 0$ $\sin KL = 0$ when $KL = n\pi$

$$\therefore k = \frac{n\pi}{L} \rightarrow ③.$$

where $k^2 = \frac{8\pi^2 m}{h^2}E$

Sub ③ in ② we get

$$\psi = A\sin\frac{n\pi}{L}x + 0\cos\frac{n\pi}{L}x$$

$$\psi_n = A \sin \frac{n\pi}{L} x \rightarrow (4)$$

Let $k^2 = \frac{8\pi^2 m}{h^2} E$

$$k^2 = \frac{n^2\pi^2}{L^2}$$

$$\frac{8\cancel{\pi^2} mE}{h^2} = \frac{n^2\cancel{\pi^2}}{L^2}.$$

$$\boxed{E_n = \frac{n^2h^2}{8ml^2}} \quad n = 1, 2 \ldots .$$

$$\boxed{E_n = \frac{n^2h^2}{8mL^2}} \rightarrow (5) \quad \text{Eigen Value.}$$

Let (4) $\psi_n = A \sin \frac{n\pi}{L} x.$

To calculate A ψ has to be normlked

$$P = \int |\psi n|^2 dx = 1$$

$$\int_0^2 |\psi_n|^2 dx = 1.$$

$$\int_0^2 |\psi_n|^2 dx = 1$$

$$\int_0^2 (A \sin \frac{n\pi}{L} x)^2 dx = 1 \qquad \left[\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)\right]$$

$$\int_0^2 [A^2 \sin^2 \frac{n\pi}{L} x] dx = 1$$

$$A^2 \int_0^2 \frac{1}{2}(1 - \cos \frac{2n\pi}{L} x) dx = 1$$

$$\therefore \frac{A^2}{2}[x]_0^L - \frac{L}{2\pi n}\left[\sin \frac{2n\pi}{L} x\right]_0^L = 1$$

$$\frac{A^2}{2}[L] - \frac{L}{2n\pi}\left[\sin \cancel{\frac{2\pi n}{L}L}^{0} - \sin \cancel{\frac{2n\pi}{L}0}^{0}\right] = 1$$

$$\frac{A^2 L}{2} = 1$$

$$A^2 = \frac{2}{L} \qquad \therefore A = \sqrt{\frac{2}{L}}$$

Substituting A in eqn ⑤ we get

$$\boxed{\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi}{L} x} \rightarrow ⑥.$$

Eigen Function.

Note:- only for info full

Interpretation of Result

$$E_n = \frac{n^2 h^2}{8mL^2}$$

$n = 1, L, = 3$

Let $n = 1$, $E = \frac{h^2}{8mL^2} = E_0 = E_0$

$n = 2$, $E = \frac{4h^2}{8mL^2} = 4E_0 = E_1$

$n = 3$, $E = \frac{9h^2}{8mL^2} = 9E_0 = E_2$

$$\boxed{\therefore E_{n-1} = n^2 E_0}$$ zero point energy

for ~~given~~ groundstate $P = \int_0^2 |\psi_2^2| dn$

$n = 1$

$\psi_1 = \sqrt{\frac{2}{L}} \sin \frac{\pi}{L} x.$

$x = 0$, $\psi_1 = \sqrt{2/L} \sin \pi/L (0) \therefore \psi_1 = 0.$

$x = \frac{L}{2}$ $\psi_1 = \sqrt{\frac{2}{L}} \sin \frac{\pi}{\cancel{L}} \times \frac{\cancel{L}}{2}$

$= \sqrt{2/L} \sin \pi/2$

$\psi_1 = \sqrt{2/L}.$

$x = L$ $\psi_1 = \sqrt{\frac{2}{L}} \sin \frac{\pi}{\cancel{L}} \cancel{L} = \sqrt{\frac{2}{L}} \sin \pi$

$\psi_1 = 0.$

$\therefore P = |\psi_1|^2$

$= \left(\sqrt{\frac{2}{L}}\right)^2.$

$= \frac{2}{L}$

For excited state

$n = 2$

$\psi_2 = \sqrt{\frac{2}{L}} \sin \frac{2\pi}{L} x.$

$x = 0$ $\psi_2 = 0.$

$x = L/4$ $\psi_2 = \sqrt{\frac{2}{L}} \sin \frac{2\pi}{L} \times \frac{L}{4}$

$= \sqrt{\frac{2}{L}} \sin \pi/2$

$\psi_2 = \sqrt{\frac{2}{L}}$

$x = \frac{L}{2}$

$\psi_2 = \sqrt{\frac{2}{L}} \sin \frac{2\pi}{L} \times \frac{L}{2}$

$= \sqrt{\frac{2}{L}} \sin \pi$

$\psi_2 = 0.$

$x = \frac{3L}{4}$ $\psi_2 = \sqrt{\frac{2}{L}} \sin \frac{2\pi}{L} \times \frac{3L}{4}$

$= \sqrt{\frac{2}{L}} \sin \frac{3\pi}{2}$ $\psi_2 = -\sqrt{\frac{2}{L}}$